SUSHREE SWATEEPRAJNYA BEHERA

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PROFESSIONAL SUMMARY

I am an experimental quantum hardware researcher with hands-on experience in designing, integrating, and characterizing complex photonic and atomic systems. I am proficient in laboratory automation, subsystem interfacing, and building precision measurement platforms from scratch for high-stability environments. Alongside, I am experienced in navigating cleanroom fabrication workflows, managing multi-instrument data acquisition pipelines, and translating high-level physics requirements into scalable quantum hardware solutions.

Research Interest: Solid-state quantum photonics, quantum light sources, nanophotonic device integration, quantum networks, and precision optical measurement platforms.

RESEARCH POSITIONS

Quantum Sensing Researcher March 2026 - July 2026
QuBeats (M/s Nostradamus Technologies Private Limited), Hyderabad, India
In collaboration with Tata Institute of Fundamental Research Hyderabad

Researcher November 2024 - December 2025
Quantum Materials Team
Luxembourg Institute of Science and Technology, Esch-sur-Alzette
In association with Université du Luxembourg, Luxembourg

Guest Scientist May 2024 - July 2024
Quantum Photonics Group - Experimental Physics
Universität des Saarlandes, Saarbrücken, Germany

Project Associate I October 2022 - April 2024
Indian Institute of Science, Bengaluru, India
Quantum Optics and Quantum Information Processing Group
Department of Instrumentation and Applied Physics
Center of Excellence in Quantum Technology

Visiting researcher July 2022 - October 2022
Department of Physics
Institute of Mathematical Sciences, Chennai, India

EDUCATION

Indian Institute of Science Education and Research Tirupati, India

BS-MS Dual Degree

August 2016 – August 2021

MS Thesis Work: "Quantum Volume: Capability of a quantum computer and effective implementation of quantum circuits"

- Computed noise in state preparation, gate operations, and measurements that limits a quantum processor performance.
- Analysed a protocol for accurate estimation of quantum volume in noisy NISQ devices.

RESEARCH PROJECTS

QuBeats (M/s Nostradamus Technologies Private Limited), India Supervisor: Dr. Shouvik Mukherjee
March 2026 – July 2026
Quantum Sensing Researcher

Warm atomic vapour experiments - optically pumped magnetometers with Rubidium (Rb) atoms

- Designed and built a GUI for compact optically pumped magnetometer to sense and measure the magnetic field of earth.
- Performed data acquisition and analysis of frequency-modulated signals in a rubidium (Rb)-based nonlinear magneto-optical rotation (NMOR) experiment using the Moku:Go platform, integrating waveform generation, lock-in detection, and oscilloscope measurements.
- Built a python-based digital lock-in amplifier for signal extraction and phase-sensitive detection using butterworth low-pass filtering.
- Designed, wound, and characterized Helmholtz coils for precision magnetic-field generation and control in quantum optics experiments.

Luxembourg Institute of Science and Technology, Luxembourg Supervisor: Dr. Florian Kaiser
November 2024 – December 2025
Researcher: Quantum Materials Team
In asssociation with Université du Luxembourg, Luxembourg

Two-photon interference using color centers in silicon carbide (4H-SiC), under the umbrella of AQuaTSiC (Advanced Quantum Technologies with Silicon Carbide) project

- Designed and built an optical fiber gluing station for waveguide–fiber interfacing from scratch, enabling stable chip–fiber coupling.
- Integrated visible to near-infrared (780–1220 nm) laser systems with 4H-SiC nanophotonic devices.
- Led cleanroom fabrication of specialty fibers (1060XP and ZBLAN in association with Flawless Photonics Inc.) and process logistics coordination, including process optimization, execution, and quality control.
- Evaluated, qualified, and procured optical adhesives for photonic chip-level fiber attachment, defining bonding protocols and equipment requirements for long-term mechanical and optical stability.
- Conducted theoretical study of neutral divacancy (VV) in 4H-SiC, analyzing electronic structure, and optical transitions involved in single-photon generation.

Universität des Saarlandes, Germany Supervisor: Prof. Jürgen Eschner
May 2024 – July 2024
Guest Scientist: Quantum Photonics Group

Lasing from Cold Ytterbium Atoms, Quantum Communication with Calcium (Ca^+) ions and Quantum Photonics

- Investigated coherent population trapping and dark-state formation in a Λ-system of ^{174}Yb using 680 nm and 770 nm repump lasers; performed fluorescence spectroscopy on the $^1S_0$ to $^3S_1$ transition via $^3P_1$.
- Built a double pass AOM (Acousto-Optic Modulator) setup for 397nm laser and data acquisition, analysis, and stability characterization of pulsed laser with varying radio-frequency using object oriented programming with MATLAB.

Indian Institute of Science, India Supervisor: Dr. C. M. Chandrashekar and Dr. K. Muhammed Shafi
October 2022 - April 2024
Project Associate I: Quantum Optics and Quantum Information Processing Group
Center of Excellence in Quantum Technology

Quantum hyperentanglement with spontaneous parametric down-conversion (SPDC) using type I and type II BBO crystals and quantum entanglement using Type II SPDC with PPKTP Crystal.

- Generation of entangled and heralded single photon sources, detection and estimation by EMCCD camera to establish the violation of Bell's inequalities.

Institute of Mathematical Sciences (IMSc), India Supervisor: Dr. C. M. Chandrashekar
July 2022 – October 2022
Visiting Researcher

- Topological networks for quantum communication between distant qubits, robust networks for efficient communication.

LABORATORY HARDWARE EXPERIENCE

Lasers & Optics: CW lasers (visible to near-infrared), free-space and fiber optics, optical components (mirrors, lenses, beam splitters, irises), AOMs, RF drivers, spectroscopy, fiber splicing and high-speed spectrometer

Detectors & Timing: Single-photon detectors (SPCM-AQ4C, APDs), Photomultiplier tube (PMT), EMCCD, time-correlated single-photon counting (TCSPC), Time Tagger S15, HYDRA II, PicoHarp, and PD10B High-gain balanced photodetector.

Automation & Electronic Instrumentations: Hardware automation and synchronization of lasers, CMOS cameras, 6-axis stages, Digital oscilloscopes (PicoScope series, Moku:Go, Digilent AD3), lock-in amplifiers, waveform generators, RF drivers, current sources and precision magnetic-field controllers (Twinleaf systems).

Cleanroom Experience (ISO 4) & Characterization: Specialty fiber fabrication (1060XP, ZBLAN), nanophotonic device processing workflows and fiber–chip integration techniques, nanostructure characterization using SEM and AFM, and laboratory safety and process protocol compliance in controlled environments.

ADDITIONAL SKILLS

Programming Language	Python
Quantum SDK	Qiskit
Software	Mathematica, MATLAB, Overleaf and LaTeX, OriginPro
Platforms	Amazon Braket (SV1 simulator, hardware backends), IBM Quantum Experience

PUBLICATIONS

Observation of geometric phase in a molecular Aharonov-Bohm system using IBM Quantum Computer
Gaurav Rudra Malik, *Sushree Swateeprajnya Behera*, Shubham Kumar, Bikash K. Behera, Prasanta K. Panigrahi
Preprint: arXiv:1909.00298 [quant-ph], 2019

Controlling asymptotic self-similar optical beams in a semiconductor waveguide doped with quantum dots
Sushree Swateeprajnya Behera, Nikhitha Nunavath, Vineetha Ashok, Thokala Soloman Raju Published as conference paper at Optical Society of America (OSA), PHOTONICS 2018

INVITED TALKS AND PRESENTATIONS

Two-photon interference with divacancies in silicon carbide
Invited talk - oral presentation
Quantum SiC Workshop, Stralsund, Germany (October 2025)

Towards two-photon interference with color centers in silicon carbide
Oral and poster presentation
Greenhorn Meeting, Saarbrücken, Germany (September 2025)

Multiplexing divacancies in silicon carbide for quantum networks
Poster presentation
DPPM PhD Day, University of Luxembourg, Luxembourg (April 2025)

Multiplexing color centers in silicon carbide for quantum networks
Poster presentation
DPG Conference, Bonn, Germany (March 2025)

Journal club presentation in quantum metrology
Oral presentation - Studied the optimal bound for quantum precision enhancement compared to the classical scenario
IISC Bengaluru, India (June 2023)

SUMMER SCHOOLS

Qubit by Qubit, Introduction to Quantum Computing Course (October 2020 - May 2021)
IBM quantum & QISKIT Community, The Coding School

Quantum Information and Quantum Technology (QIQT-2019)
Summer School, IISER Kolkata

ACADEMIC ACHIEVEMENTS

Graduate Aptitude Test in Engineering Physics GATE 2023
Qualified
Dept. of Science and Technology, Govt. of India

Winner at Poster presentation, International Conference on Fibre Optics and Photonics
(PHOTONICS 2018) IIT Delhi, India

ADDITIONAL EXPERIENCE

Open Day 2024: IISc Bengaluru
Volunteer Research Assistant: Laboratory Tour
Indian Institute of Science (IISC) Bengaluru, India

PHOTONICS 2023: International conference on fibre optics and photonics
Indian Institute of Science (IISC) Bengaluru, India
Conference Volunteer: Laboratory Tour

Open Day 2023: IISc Bengaluru
Volunteer Research Assistant: Laboratory Tour
Indian Institute of Science (IISC) Bengaluru, India

Students Conference on Photonics (SCOP) - 2022
PRL (Physical Research Laboratory), Ahmedabad, India

Certifications:

Fundamentals of Quantum Algorithms · Basics of Quantum Information · AWS Knowledge: Amazon Braket · Integrating Quantum and HPC · Quantum Diagonalization Algorithms · Variational Algorithm Design · Quantum Machine Learning · Foundations of Quantum Error Correction · General Formulation of Quantum Information · Quantum-Safe Cryptography · Quantum Business Foundations